# Week 8: Functions+Midsem week

# Lab Exam announcements

- Lab exam tomorrow (19th Sept, Sat) will be held in 3 batches:
  - Batch 1: (9am-11am)
    - EE + BT+ CH+ CO
  - Batch 2: (11:30am-1:30pm)
    - BM+CS+ES+IC+MA
  - Batch 3: (2:00pm-4pm)
    - AI + CE + EP+ ME
- Venue: **CS109, CS209, MAE CAE Lab**
- Detailed seating arrangement to be emailed shortly.
- Follow the seating arrangement strictly.

# Lab exam guidelines

- Code to be written and submitted on Prutor only **via the desktop machines in the lab.** By now, you should be familiar with the Prutor interface.
- Make sure you remember your password. Resets cannot be done during the exam.
- No other electronic devices allowed during the exam.
- Prutor access from outside the lab will be **disabled at 4pm today**. If you want to reset your passwor

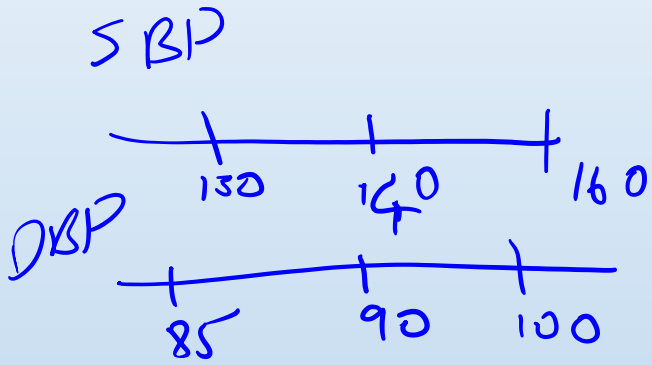
# Practice Problem 2

## Question

Read systolic blood pressure ( `SBP` ) and diastolic blood pressure ( `DBP` ) and classify them as follows:

- **Optimal:** `SBP < 130` and `DBP < 85`
- **High-normal:** `SBP` is 130–139 or `DBP` is 85–89
- **Stage 1 hypertension:** `SBP` is 140–159 or `DBP` is 90–99
- **Stage 2 hypertension:** `SBP >= 160` or `DBP >= 100`

If the two measurements indicate different categories, print the more severe category.




SBP = 135
DBP = 91

Check conditions in order of decreasing severity.

# Notes on solution to previous

- Previous question: Check the conditions in order of severity (most severe first) when using if .. Else –if statement.

- The reason is that we need to print the most severe condition. A set of values (sbp, dbp) may pass multiple conditions.

- Extension: Given a set of readings of patients, find the number of patients with each given condition.

# Practice Problem 5

**Question**

A rectangular region consists of integer grid points. The elevation at point $(x, y)$ is

$$h(x, y) = 20 + x + 2y.$$

Given `m`, `n`, and the flood-water level `H`, consider every grid point satisfying

$$0 \le x \le m, \qquad 0 \le y \le n.$$

A point is flooded if its elevation is strictly less than `H`. Count the number of flooded grid points.

**Input**

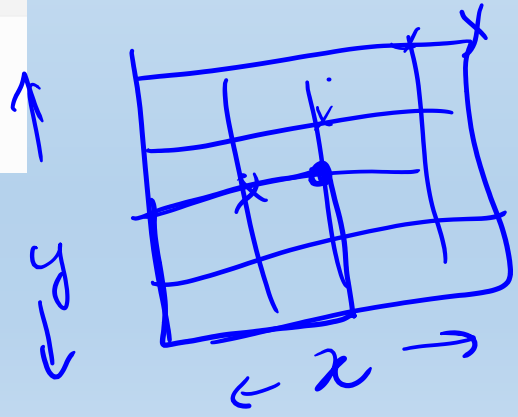
```
m n H
```

**Output**

Print the number of grid points with elevation less than `H`.

if $h(x, y) < H$ flooded else not flooded

o/p which y-coord has max #1 flooded points

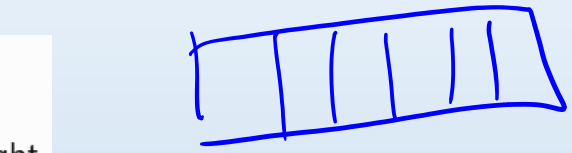


- Set up a nested while loop and check the condition at every point of the grid.

- Important: Check boundary conditions so that you don’t miss the grid endpoints (the grid is of size (m+1)(n+1) points, not m n, since indices begin at 0)

- Extension: Find the y-coordinate with the maximum number of flooded points

# Practice Problem 6



### Question

A word is called **alphabetical** if its letters occur in nondecreasing alphabetical order from left to right. Repeated letters are allowed.

For example, `aabbcz` is alphabetical.

Read a lowercase word and print `YES` if it is alphabetical. Otherwise, print `NO`.

### Examples

```
Input
almost

Output
YES
```

```
Input
apple

Output
NO
```